Micro-Level Social Network Analysis

Friend Recommendation System — Project Report

1. Network Overview

Metric	Value
Total Users (Nodes)	8
Total Friendships (Edges)	11
Network Density	0.3929
Average Degree	2.75

The network consists of 8 users connected by 11 friendship edges. A density of 0.39 indicates a moderately connected network where users are neither fully isolated nor fully interconnected.

2. Degree Centrality

Degree centrality measures how many direct connections each user has. A higher score means the user is more influential in the network.

User	Degree Centrality
Alice	0.4286
Bob	0.4286
Charlie	0.4286
David	0.4286
Eve	0.4286
Frank	0.4286
Grace	0.2857
Henry	0.2857

3. Clustering Coefficient

The clustering coefficient measures how tightly knit a user's friends are. A score of 1.0 means all of a user's friends are also friends with each other.

User	Clustering Coefficient
Alice	0.3333
Bob	0.3333
Charlie	0.3333
David	0.0000

Eve	0.0000
Frank	0.0000
Grace	0.0000
Henry	0.0000

4. Shortest Paths Between Users

Shortest path shows the minimum number of hops needed to connect two users, reflecting how closely related they are in the network.

User A	User B	Hops	Path
Alice	Bob	1	Alice → Bob
Alice	Charlie	1	Alice → Charlie
Alice	David	1	Alice → David
Alice	Eve	2	Alice → Bob → Eve
Alice	Frank	2	Alice → Charlie → Frank
Alice	Grace	2	Alice → David → Grace
Alice	Henry	3	Alice → Bob → Eve → Henry
Bob	Charlie	1	Bob → Charlie
Bob	David	2	Bob → Alice → David
Bob	Eve	1	Bob → Eve
Bob	Frank	2	Bob → Charlie → Frank
Bob	Grace	3	Bob → Alice → David → Grace
Bob	Henry	2	Bob → Eve → Henry
Charlie	David	2	Charlie → Alice → David
Charlie	Eve	2	Charlie → Bob → Eve
Charlie	Frank	1	Charlie → Frank
Charlie	Grace	3	Charlie → Alice → David → Grace
Charlie	Henry	3	Charlie → Bob → Eve → Henry
David	Eve	2	David → Frank → Eve
David	Frank	1	David → Frank
David	Grace	1	David → Grace
David	Henry	2	David → Grace → Henry
Eve	Frank	1	Eve → Frank
Eve	Grace	2	Eve → Henry → Grace

Eve	Henry	1	Eve → Henry
Frank	Grace	2	Frank → David → Grace
Frank	Henry	2	Frank → Eve → Henry
Grace	Henry	1	Grace → Henry

5. Community Detection

Using the Greedy Modularity algorithm, users are grouped into natural communities based on the density of their connections.

Group	Members
Group 1	David, Eve, Frank, Grace, Henry
Group 2	Alice, Bob, Charlie

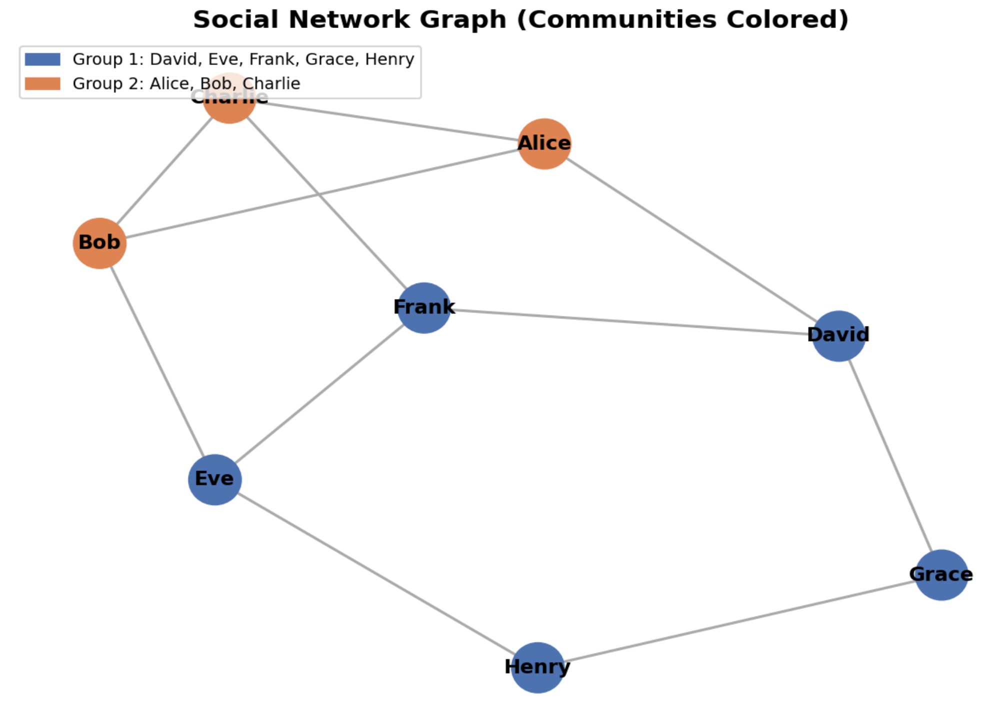
6. Friend Recommendations

Recommendations are based on mutual friend count. Users with more mutual friends are ranked higher as potential connections.

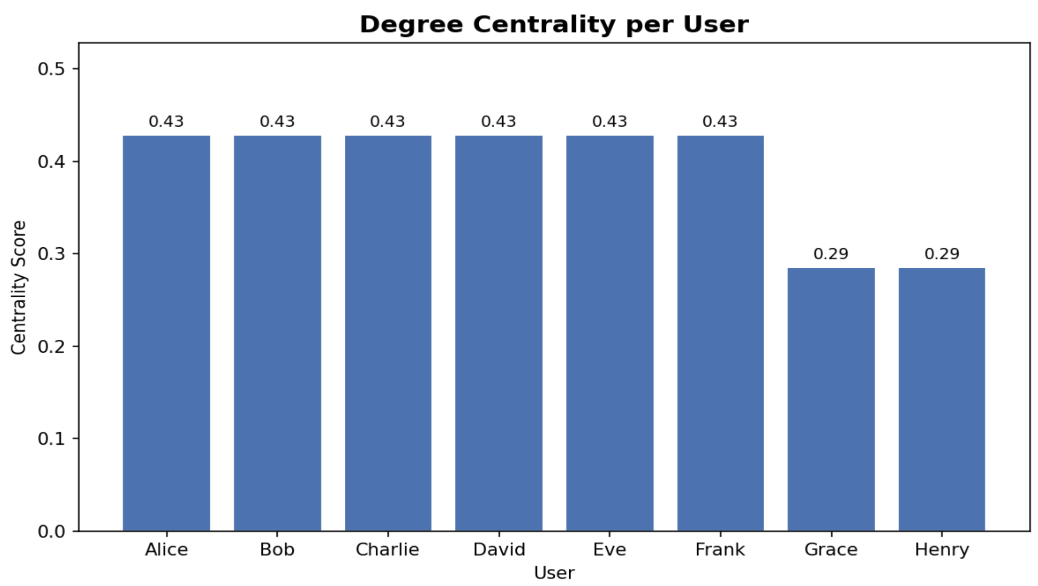
User	Recommended Friend	Mutual Friends
Alice	Frank	2
Alice	Eve	1
Bob	Frank	2
Bob	David	1
Charlie	David	2
Charlie	Eve	2
David	Charlie	2
David	Bob	1
Eve	Charlie	2
Eve	Grace	1
Frank	Alice	2
Frank	Bob	2
Grace	Eve	1
Grace	Alice	1
Henry	David	1
Henry	Bob	1

7. Visualizations

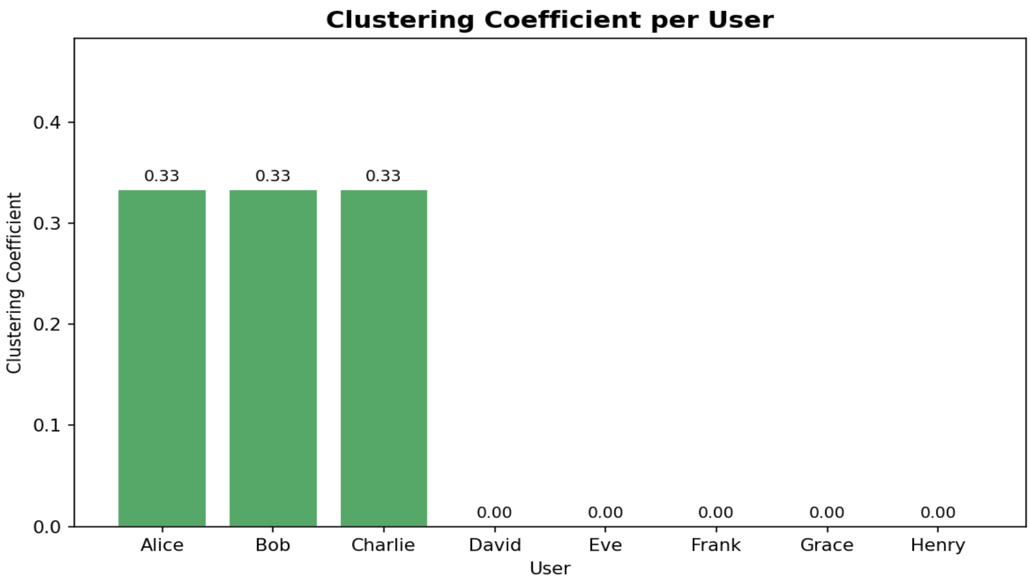
Social Network Graph (Communities Colored):



Degree Centrality per User:



Clustering Coefficient per User:



Clustering Coefficient per User
Clustering Coefficient
0.0
0.1
0.2
0.3
0.4
0.33
0.33
0.33
0.00
0.00
0.00
0.00
0.00
Alice
Bob
Charlie
David
Eve
Frank
Grace
Henry
User

8. Conclusion

This micro-level social network analysis examined a small network of 8 users. Key findings include: Alice, Bob, Charlie, David, Eve, and Frank are the most central users with equal degree centrality scores. Alice, Bob, and Charlie form tighter friend clusters compared to others. Community detection revealed two natural groups — one centered around Alice/Bob/Charlie and another around David/Eve/Frank/Grace/Henry. The friend recommendation system successfully identified meaningful connections based on mutual friend overlap, which can scale to larger real-world networks.